



FISHER INFORMATION





Notably when $f \equiv 1$ we have

$$\pi_{\beta}[\dot{\pi}_{\beta}] = \frac{\mathrm{d}}{\mathrm{d}\beta} \pi_{\beta}[1] = \frac{\mathrm{d}}{\mathrm{d}\beta} 1 = 0$$

$$I(\beta) \coloneqq \sqrt{\beta} [\dot{\pi}_\beta] = \pi_\beta [\dot{\pi}_\beta^2]$$

► Measures how sensitive the annealing path is to small changes in β

► We can measure the magnitude of the change using the variance deviation of $\dot{\pi}_\beta$ at β



The percentage change over ~~X~~ is zero as $\Delta\beta \rightarrow 0$

$$\frac{\pi_{\beta'}(x)}{\pi_{\beta}(x)} = 1 + \Delta\beta\dot{\pi}_{\beta}(x) + O(\Delta\beta^2)$$

▶ The velocity $\dot{x}_\beta(x)$ measures the percentage change in mass at x from β to $\beta + \Delta\beta$

► This coincides with **Fisher information** as the variance at β

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- ▶ Noteably when $f = 1$ we have

$$\pi_\beta[\dot{\pi}_\beta] = \frac{d}{d\beta} \pi_\beta[1] = \frac{d}{d\beta} 1 = 0$$

- ▶ The percentage change over $\mathbb{X}$ is zero as $\Delta\beta \rightarrow 0$
- ▶ We can measure the magnitude of the change using the variance deviation of $\dot{\pi}_\beta$ at β
 - ▶ This coincides with **Fisher information** as the variance at β

$$I(\beta) := \mathbb{V}_\beta[\dot{\pi}_\beta] = \pi_\beta[\dot{\pi}_\beta^2]$$

- ▶ Measures how sensitive the annealing path is to small changes in β

EXAMPLE GAUSSIAN PATHS